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United States Patent	12416876
Kind Code	B2
Date of Patent	September 16, 2025
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Structure and image forming apparatus

Abstract

Each of four outer side surfaces of each of a plurality of square pipes include a groove portion formed along a longitudinal direction of each of the plurality of square pipes and a pair of ridge line portions forming edge portions on both sides of the groove portion. An additional member includes a contact plane which is a plane in contact with a portion of the plurality of square pipes. The additional member is fixed to a plurality of first target pipes which are some of the plurality of square pipes with the contact plane in contact with at least one of the pair of ridge line portions, or the additional member is fixed to a second target pipe which is one of the plurality of square pipes with the contact plane in contact with the pair of ridge line portions.

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Appl. No.: 18/735708

Filed: June 06, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240411238 A1	Dec. 12, 2024

Foreign Application Priority Data

JP	2023-094456	Jun. 08, 2023
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Publication Classification

Int. Cl.: G03G15/01 (20060101); G03G15/20 (20060101)

U.S. Cl.:

CPC G03G15/0142 (20130101); G03G15/2064 (20130101);

Field of Classification Search

CPC: H04N (1/40037)

USPC: 399/110

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5423604	12/1994	Chern	312/265.5	A47B 47/005
6259872	12/2000	Fukunaga	399/110	G03G 15/00

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2000138470	12/1999	JP	N/A
3566115	12/2003	JP	N/A

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Background/Summary

INCORPORATION BY REFERENCE

(1) This application is based upon and claims the benefit of priority from the corresponding Japanese Patent Application No. 2023-094456 filed on Jun. 8, 2023, the entire contents of which are incorporated herein by reference.

BACKGROUND

(2) The present disclosure relates to a structure including a plurality of joined square pipes, and an image forming apparatus including the structure.

(3) The image forming apparatus includes a sheet feed portion, a printing portion, a developer container, and the like. The image forming apparatus further includes a structure that supports the sheet feed portion, the printing portion, and the developer container.

(4) For example, it is known that the structure includes a plurality of square pipes joined by welding. The structure having such a configuration has high strength.

SUMMARY

(5) A structure according to an aspect of the present disclosure includes a plurality of joined square pipes and an additional member attached to a portion of the plurality of square pipes. Four corner portions of each of the plurality of square pipes include three first corner portions formed by bending a single metal plate, and one second corner including two edge portions of the metal plate and a welded portion where the two edge portions are joined. Each of four outer side surfaces of each of the plurality of square pipes includes a groove portion formed along a longitudinal

direction of each of the plurality of square pipes and a pair of ridge line portions forming edge portions on both sides of the groove portion. The additional member has a contact plane which is a plane in contact with a portion of the plurality of square pipes. The additional member is fixed to the plurality of first target pipes with the contact plane in contact with at least one of the pair of ridge portions in each of the plurality of first target pipes which are some of the plurality of square pipes, or the additional member is fixed to the second target pipe with the contact plane in contact with the pair of ridge portions in the second target pipe which is one of the plurality of square pipes.

(6) An image forming apparatus according to another aspect of the present disclosure includes the structure and a printing portion mounted on the structure and configured to form an image on a sheet.

(7) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description with reference where appropriate to the accompanying drawings. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Furthermore, the claimed subject matter is not limited to implementations that solve any or all disadvantages noted in any part of this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a configuration diagram of an image forming apparatus including a structure according to an embodiment.

(2) FIG. 2 is a perspective view of the structure according to the embodiment.

(3) FIG. 3 is a perspective view of one of square pipes in the structure according to the embodiment.

(4) FIG. 4 shows a part of a manufacturing process of the square pipes in the structure according to the embodiment.

(5) FIG. 5 is a perspective view of a plurality of first target pipes and a first additional member in the structure according to the embodiment.

(6) FIG. 6 is a perspective view of a second target pipe and a second additional member in the structure according to the embodiment.

DETAILED DESCRIPTION

(7) Hereinafter, an embodiment of the present disclosure will be described with reference to the drawings. It is noted that the following embodiment is an example of embodying the present disclosure and does not limit the technical scope of the present disclosure.

(8) [Configuration of Image Forming Apparatus 10]

(9) A structure 6 according to the embodiment is employed in an image forming apparatus 10. As shown in FIG. 1, the image forming apparatus 10 includes a main body unit 1 and an image reading unit 2.

(10) The main body unit 1 includes a sheet feed portion 30, a paper sheet conveying device 3, a printing portion 4, one or more developer containers 5, and a structure 6. The image reading unit 2 is connected to the top of the main body unit 1. The image reading unit 2 is a device that reads an image of a document sheet.

(11) The sheet feed portion 30 is a device that stores a plurality of paper sheets and feeds the stored paper sheets one by one to a paper sheet conveying path 300. The paper sheet conveying device 3 includes a plurality of conveying roller pairs 31 that convey the paper sheet along the paper sheet conveying path 300.

(12) The printing portion 4 is a device that forms an image on the paper sheet supplied from the

sheet feed portion **30** through the paper sheet conveying device **3**. In the example shown in FIG. **1**, the printing portion **4** forms an image on the paper sheet using an electrophotographic method. The paper sheet is an example of the sheet.

(13) The electrophotographic printing portion **4** includes a laser scanning unit **40**, one or more image forming portions **4x**, a transfer device **44**, and a fixing device **45**.

(14) In the example shown in FIG. **1**, the printing portion **4** is a tandem type color image printing device. Therefore, the printing portion **4** includes four image forming portions **4x** corresponding to four color toners of yellow, cyan, magenta, and black. The image forming apparatus **10** further includes four developer containers **5** corresponding to the four image forming portions **4x**.

(15) In each of the image forming portions **4x**, a drum-shaped photoconductor **41** rotates, and a charging device **42** charges the outer peripheral surface of the photoconductor **41**. Further, the laser scanning unit **40** writes an electrostatic latent image on the outer peripheral surface of the photoconductor **41**, and a developing device **43** develops the electrostatic latent image into a toner image.

(16) Furthermore, in the transfer device **44**, an intermediate transfer belt rotates in contact with the four photoconductors **41**, and four primary transfer devices corresponding to the four image forming portions **4x** each transfer the toner image onto the intermediate transfer belt. Further, a secondary transfer device transfers the toner image on the intermediate transfer belt to the paper sheet being conveyed along the paper sheet conveying path **300**.

(17) The fixing device **45** heats and presses the toner image on the paper sheet to fix the toner image on the paper sheet. The paper sheet conveying device **3** discharges the paper sheet with the image formed thereon from the paper sheet conveying path **300**.

(18) Each of the developer containers **5** supplies toner to the corresponding developing device **43** in the printing portion **4**. The toner is an example of the developer. It is noted that the printing portion **4** may be a device that forms an image on the paper sheet by an inkjet method or another method.

(19) The sheet feed portion **30**, the paper sheet conveying device **3**, the printing portion **4**, and the developer containers **5** are attached to the structure **6**. The image reading unit **2** is connected to the top of the structure **6**.

(20) The structure **6** includes a plurality of joined square pipes **60** (see FIG. **2**). In the structure **6**, a plurality of metal square pipes **60** are joined by welding. Such a structure **6** has high strength.

(21) By the way, a general square pipe is manufactured by bending a single metal plate and a welding process of joining two edge portions of the metal plate together.

(22) On the other hand, one or more additional members **7** may be attached to a portion of the plurality of square pipes **60** along an outer side surface of the portion of the plurality of square pipes **60** (see FIG. **2**). In this case, it may be required that the additional members **7** are attached with high positional accuracy.

(23) For example, the additional member **7** is a sheet metal member manufactured by one or both of bending and drilling of a metal plate.

(24) In the present embodiment, the structure **6** includes a plurality of additional members **7** each attached to a portion of the plurality of square pipes **60** (see FIG. **2**). The additional members **7** are each fixed to a portion of the plurality of square pipes **60** by welding or screws.

(25) In the conventional structure, it was difficult to attach the additional member **7** to a portion of a plurality of conventional square pipes with high positional accuracy due to insufficient accuracy of the shape of the outer surface of each of the plurality of conventional square pipes.

(26) In the present embodiment, the structure **6** is configured such that the additional member **7** can be attached to the outer side surface of a portion of the plurality of square pipes **60** with high positional accuracy. Hereinafter, the configuration will be described.

(27) [Configuration of Each Square Pipe **60**]

(28) The plurality of square pipes **60** each have four outer side surfaces **62** and four corner portions **63** (see FIG. **3**). The four outer side surfaces **62** and the four corner portions **63** are formed to

extend in the longitudinal direction of each of the square pipes **60**.

(29) The four corner portions **63** include three first corner portions **63a** and one second corner portion **63b**. The three first corner portions **63a** are three bent portions formed by bending a single metal plate. The second corner portion **63b** includes two edge portions **61** of the metal plate and a welded portion **600** where the two edge portions **61** are joined.

(30) The two edge portions **61** and the welded portion **600** are formed to extend in the longitudinal direction of each of the square pipes **60**.

(31) Each of the square pipes **60** has a plurality of through-holes **64** formed in one or more of the four outer side surfaces **62** (see FIG. 3). The processing of the plurality of through-holes **64** in the plurality of square pipes **60** is performed on a flat metal plate before being bent.

(32) Some or all of the plurality of through-holes **64** are used for mounting components. For example, each component is directly mounted on one of the outer side surfaces **62** of each of the square pipes **60** by fixing members, such as screws, which are inserted into the through-holes **64**.

(33) In the present embodiment, the image reading unit **2**, the paper sheet conveying device **3**, and the printing portion **4** are mounted on the structure **6** by the fixing members inserted into the through-holes **64**.

(34) It is conceivable that two or more of the plurality of square pipes **60** may be of the same type of member for commonality of components. In this case, some of the plurality of through-holes **64** may not be used for mounting components in some of the plurality of square pipes **60** of the same type.

(35) In each of the square pipes **60**, none of the four outer side surfaces **62** includes the welded portion **600**. Therefore, it is easy to form one or more through-holes **64** in each of the four outer side surfaces **62** of each of the square pipes **60**.

(36) For example, four or more through-holes **64** are formed in the four outer side surfaces **62** of at least one of the plurality of square pipes **60** (see FIG. 3).

(37) In each of the square pipes **60**, two out of the four outer side surfaces **62** are formed between the three first corner portions **63a**. The remaining two of the four outer side surfaces **62** are formed between two of the three first corner portions **63a** and the second corner portion **63b**.

(38) Since the square pipe **60** as shown in FIG. 3 is employed in the structure **6**, all of the four outer side surfaces **62** of each of the square pipes **60** can be used for mounting components.

(39) In the present embodiment, the three first corner portions **63a** have a chamfered shape (see FIG. 3). In addition, the two edge portions **61** of the metal plate in the second corner portion **63b** are bent to form a chamfered shape.

(40) In each of the square pipes **60**, the welded portion **600** may be formed so as to slightly protrude from the gap between the two edge portions **61**. Even in such a case, the two edge portions **61** have a chamfered shape, so that the welded portion **600** does not interfere with the component mounted on each of the two outer side surfaces **62** adjacent to the second corner portion **63b**.

(41) Each of the four outer side surfaces **62** of each of the plurality of square pipes **60** has a groove portion **62a** and a pair of ridge line portions **62b** (see FIG. 3 and FIG. 4).

(42) The groove portion **62a** is a concave portion formed along the longitudinal direction of each of the plurality of square pipes **60**. The pair of ridge line portions **62b** form the edge portions on both sides of the groove portion **62a**. The pair of ridge line portions **62b** are also formed along the longitudinal direction of each of the plurality of square pipes **60**.

(43) The pair of ridge line portions **62b** form a pair of top portions between a pair of step portions and a pair of corner portions **63** on both sides of the groove portion **62a**. Each of the plurality of square pipes **60** has four groove portions **62a** and four pairs of ridge line portions **62b** corresponding to the four outer side surfaces **62**.

(44) For example, the step between each groove portion **62a** and the pair of ridge line portions **62b** on both sides of the groove portion **62a** is about 0.3 mm to 0.8 mm.

(45) In the following description, the square pipe before the groove portions **62a** and the pair of

ridge line portions **62b** of the four outer side surfaces **62** are formed will be referred to as an original square pipe **60P** (see FIG. 4).

(46) As shown in FIG. 4, the groove portion **62a** and the pair of ridge line portions **62b** of each of the four outer side surfaces **62** of each of the plurality of square pipes **60** are formed by a roll press in which the original square pipe **60P** is passed between four pressure rollers **8**.

(47) Two of the four pressure rollers **8** and the remaining two pressure rollers **8** are arranged to face each other with a molding space **80** into which the original square pipe **60P** is inserted interposed therebetween. The four pressure rollers **8** are fixed at predetermined positions.

(48) When the original square pipe **60P** is passed through the molding space **80**, the four pressure rollers **8** rotate while making pressure contact with the four outer side surfaces of the original square pipe **60P**. Thus, four groove portions **62a** corresponding to the surface shapes of the four pressure rollers **8** are formed on the four outer side surfaces of the original square pipe **60P**.

(49) In addition, at the same time that the four groove portions **62a** are formed, a pair of ridge line portions **62b** are formed on both sides of each of the four groove portions **62a**.

(50) Further, when the roll press is performed, the inner side surfaces of the original square pipe **60P** are not restrained. Therefore, the four inner side surfaces **65** of each of the plurality of square pipes **60** are formed so as to convex inward, corresponding to the four groove portions **62a** (see FIG. 3 and FIG. 4).

(51) Even when the dimensional accuracy of the four outer side surfaces of the original square pipe **60P** is not sufficient, the dimensional accuracy of the four groove portions **62a** and the four pairs of ridge line portions **62b** in the square pipe **60** after molding is high.

(52) On the other hand, each of the additional members **7** has a contact plane **7a** which is a plane in contact with a portion of the plurality of square pipes **60** of the structure **6** (see FIG. 5 and FIG. 6). Hereinafter, two types of members of the additional members **7** will be referred to as a first additional member **7x** and a second additional member **7y**, respectively (see FIG. 2, FIG. 5, and FIG. 6).

(53) Among the plurality of square pipes **60**, a plurality of pipes to which the first additional member **7x** is attached will be referred to as first target pipes **60x** (see FIG. 5). Similarly, among the plurality of square pipes **60**, a pipe to which the second additional member **7y** is attached will be referred to as a second target pipe **60y** (see FIG. 6).

(54) The first additional member **7x** is fixed to the plurality of first target pipes **60x** with the contact plane **7a** of the first additional member **7x** in contact with at least one of the pair of ridge line portions **62b** of each of the plurality of first target pipes **60x** (see FIG. 5).

(55) In the example shown in FIG. 5, a plurality of fixing portions **71** of the first additional member **7x** are welded to the groove portion **62a** of each of the plurality of first target pipes **60x**.

(56) When the contact plane **7a** is in contact with at least one of the pair of ridge line portions **62b** of each of the plurality of first target pipes **60x**, a slight gap is formed between the first additional member **7x** and the groove portion **62a**. When two members are arranged with a gap of about 0.3 to 0.8 millimeters therebetween, there is no particular problem in welding the two members.

(57) On the other hand, the second additional member **7y** is fixed to the second target pipe **60y** with the contact plane **7a** of the second additional member **7y** in contact with the pair of ridge line portions **62b** of the second target pipe **60y** (see FIG. 6).

(58) In the example shown in FIG. 6, the second additional member **7y** is fixed to the second target pipe **60y** by a plurality of screws **72**. The plurality of screws **72** are inserted into the plurality of through-holes **73** formed in the second additional member **7y** and the plurality of through-holes **64** formed in the second target pipe **60y** (see FIG. 6).

(59) It is noted that a plurality of fixing portions of the second additional member **7y** may be welded to the groove portion **62a** of the second target pipe **60y** (not shown).

(60) The additional member **7** is positioned with high accuracy with the contact plane **7a** of the additional member **7** in contact with at least one of the pair of ridge line portions **62b** of a portion

of the plurality of square pipes **60**.

(61) By employing the structure **6**, the additional member **7** can be mounted on the outer side surface **62** of a portion of the plurality of square pipes **60** with high positional accuracy.

(62) It is to be understood that the embodiments herein are illustrative and not restrictive, since the scope of the disclosure is defined by the appended claims rather than by the description preceding them, and all changes that fall within metes and bounds of the claims, or equivalence of such metes and bounds thereof are therefore intended to be embraced by the claims.

Claims

1. A structure including a plurality of joined square pipes and an additional member attached to a portion of the plurality of square pipes, wherein four corner portions of each of the plurality of square pipes include three first corner portions formed by bending a single metal plate, and one second corner portion including two edge portions of the metal plate and a welded portion where the two edge portions are joined, each of four outer side surfaces of each of the plurality of square pipes include a groove portion formed along a longitudinal direction of each of the plurality of square pipes and a pair of ridge line portions forming edge portions on both sides of the groove portion, the additional member includes a contact plane which is a plane in contact with the portion of the plurality of square pipes, the additional member is fixed to a plurality of first target pipes which are some of the plurality of square pipes with the contact plane in contact with at least one of the pair of ridge line portions of each of the plurality of first target pipes, or the additional member is fixed to a second target pipe which is one of the plurality of square pipes with the contact plane in contact with the pair of ridge line portions of the second target pipe, and the additional member is welded to the groove portion of each of the plurality of first target pipes or the groove portion of the second target pipe.
 2. A structure including a plurality of joined square pipes and an additional member attached to a portion of the plurality of square pipes, wherein four corner portions of each of the plurality of square pipes include three first corner portions formed by bending a single metal plate, and one second corner portion including two edge portions of the metal plate and a welded portion where the two edge portions are joined, each of four outer side surfaces of each of the plurality of square pipes include a groove portion formed along a longitudinal direction of each of the plurality of square pipes and a pair of ridge line portions forming edge portions on both sides of the groove portion, the additional member includes a contact plane which is a plane in contact with the portion of the plurality of square pipes, the additional member is fixed to a plurality of first target pipes which are some of the plurality of square pipes with the contact plane in contact with at least one of the pair of ridge line portions of each of the plurality of first target pipes, or the additional member is fixed to a second target pipe which is one of the plurality of square pipes with the contact plane in contact with the pair of ridge line portions of the second target pipe, and the groove portion and the pair of ridge line portions of each of the four outer side surfaces of each of the plurality of square pipes are formed by a roll press in which an original square pipe before the groove portion and the pair of ridge line portions of each of the four outer side surfaces are formed is passed between four pressure rollers configured to make pressure contact with four outer side surfaces of the original square pipe.
 3. An image forming apparatus comprising: the structure according to claim 1; and a printing portion mounted on the structure and configured to form an image on a sheet.
 4. An image forming apparatus comprising: the structure according to claim 2; and a printing portion mounted on the structure and configured to form an image on a sheet.
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